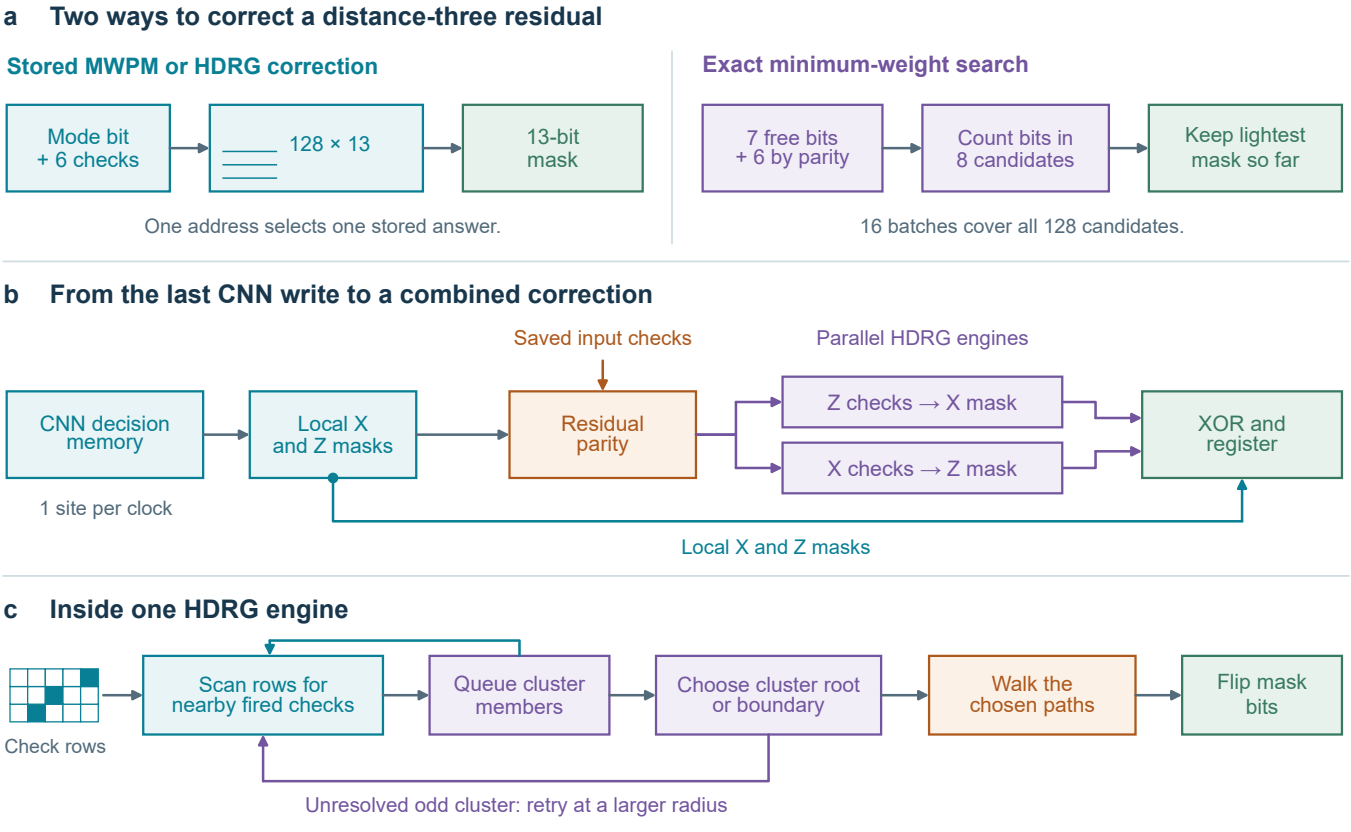

a
Two ways to correct a distance-three residual
Stored MWPM or HDRG correction
Mode bit
+ 6 checks
128 × 13
13-bit
mask
One address selects one stored answer.
Exact minimum-weight search
7 free bits
+ 6 by parity
Count bits in
8 candidates
Keep lightest
mask so far
16 batches cover all 128 candidates.
b
From the last CNN write to a combined correction
Saved input checks
Parallel HDRG engines
CNN decision
memory
Local X
and Z masks
Residual
parity
Z checks → X mask
X checks → Z mask
XOR and
register
1 site per clock
Local X and Z masks
c
Inside one HDRG engine
Check rows
Scan rows for
nearby fired checks
Queue cluster
members
Choose cluster root
or boundary
Walk the
chosen paths
Flip mask
bits
Unresolved odd cluster: retry at a larger radius